

Calcul numeric

Capitolul 3

Problema 3.9

(a) Utilizați interpolarea Hermite pentru a găsi un polinom de grad minim ce satisface

$$p(-1)=p'(-1)=0, p(0)=1, p(1)=p'(1)=0 \quad .$$

Simplificați expresia lui p cât mai mult posibil.

$$x_0=-1, x_1=0, x_2=1$$

$$r_0=1, r_1=0, r_2=1$$

$$m=2$$

$$n=m+r_0+r_1+r_2=4 \Rightarrow \text{Hf are gradul} \leq 4$$

$$(H_n f)(x) = \sum_{i=0}^m \sum_{l=0}^{r_i} f^{(l)}(x_i) * h_{il}(x)$$

$$q_i(x) = \prod_{j=0, j \neq i}^m \left(\frac{x-x_i}{x_i-x_j} \right)^{r_j+1}$$

$$h_{ir_i}(x) = \frac{(x-x_i)^{r_i}}{r_i!} * q_i(x)$$

$$h_{il}(x) = \frac{(x-x_i)^l}{l!} * q_i(x) - \sum_{j=l+1}^{r_i} \binom{j}{l} * q_i^{j-l}(x_i) * h_{ij}(x)$$

$$q_0(x) = \prod_{j=0, j \neq 0}^2 \left(\frac{x-x_i}{x_0-x_j} \right)^{r_j+1} = -\frac{x(x-1)^2}{4}$$

$$q_1(x) = \prod_{j=0, j \neq 1}^2 \left(\frac{x-x_j}{x_1-x_j} \right)^{r_j+1} = (x-1)^2 * (x+1)^2$$

$$q_2(x) = \prod_{j=0, j \neq 2}^2 \left(\frac{x-x_j}{x_2-x_j} \right)^{r_j+1} = \frac{x*(x+1)^2}{4}$$

$$h_{01} = \frac{(x-x_0)^{r_0}}{r_0!} = -\frac{x(x+1)(x-1)^2}{4}$$

$$h_{10} = \frac{(x-x_1)^{r_1}}{r_1!} * q_1(x) = \frac{x(x-1)(x+1)^2}{4}$$

$$h_{00} = \frac{(x-x_0)^0}{0!} * q_0(x) - \sum_{j=1}^{r_0} \binom{j}{0} * q_0^j(x_0) * h_{0j}(x) = q_0(x) - h_{01}(x)$$

$$h_{20}(x) = \frac{(x-x_2)^0}{0!} * q_2(x) - \sum_{j=1}^{r_2} \binom{j}{0} * q_2^j(x_2) * h_{2j}(x) = q_2(x) - h_{21}(x)$$

$$(Hf)(x) = f(x_0) * h_{00}(x) + f'(x_0) * h_{01}(x) + f(x_1) * h_{10}(x) + f(x_2) * h_{20}(x) + f'(x_2) * h_{21}(x) = \\ = (x-1)^2(x+1)^2 = x^4 - 2x^2 + 1$$

(b) Presupunem că polinomul p de la (a) este utilizat pentru a aproxima funcția $f(x) = \left[\cos\left(\frac{\pi x}{2}\right) \right]^2$ pe $-1 \leq x \leq 1$.

(b1) Exprimați eroarea $(Rf)(x) = f(x) - p(x)$ (pentru un $x \in [-1, 1]$ fixat) în funcție de o derivată corespunzătoare a lui f .

$$(R_n f)(x) = \frac{1}{n+1!} * u_n(x) f^{(n+1)}(\xi_x), \xi_x \in (a, b)$$

$$u_n(x) = \prod_{i=0}^m (x-x_i)^{r_i+1}$$

$$u_4(x) = x(x+1)^2(x-1)^2$$

$$(Rf)(x) = \frac{1}{5!} * x(x+1)^2(x-1)^2 * f^{(5)}(\xi_x)$$

$$f^{(5)}(x) = -\frac{\pi^5}{2} * \sin(\pi x) \Rightarrow (Rf)(x) = -\frac{\pi^5}{240} * x(x+1)^2(x-1)^2 \sin(\pi \xi_x)$$

$$\text{pentru } x = \frac{1}{2} \Rightarrow (Rf)(x) = \frac{3 * \pi^5}{2560} * \sin(\pi \xi_{\frac{1}{2}})$$

(b2) Găsiți o margine superioară a lui $|(Rf)(x)|$ (pentru un $x \in [-1, 1]$ fixat).

$$|(R_n f)(x)| \leq \frac{M_{n+1}}{n+1!} * |u_n(x)|$$

$$M_{n+1} = \max$$

$$u_n(x) = \prod_{i=0}^m (x - x_i)^{r_i+1}$$

$$M_5 = \max_{x \in [-1, 1]} \left| -\frac{\pi^5}{2} * \sin(\pi x) \right| = \frac{\pi^5}{2}$$

$$u_4(x) = x(x+1)^2(x-1)^2$$

$$|(Rf)(x)| \leq \frac{\pi^5}{240} * |x|(x+1)^2(x-1)^2$$

$$\text{pentru } x = \frac{1}{2} \Rightarrow \left| (Rf)\left(\frac{1}{2}\right) \right| \leq 3 \frac{\pi^5}{2560} \simeq 0,3586$$

(b3) Estimați $\max_{-1 \leq x \leq 1} |(Rf)(x)|$.

$$\max_{x \in [a, b]} |(R_n f)(x)| \leq \frac{(b-a)^{n+1}}{n+1!} * M_{n+1}$$

$$M_5 = \frac{\pi^5}{2}$$

$$\max_{x \in [-1,1]} |(Rf)(x)| \leq \frac{2^5}{120} * \frac{\pi^5}{2} \simeq 40,8026$$